

ECE 253 – Digital Image Processing

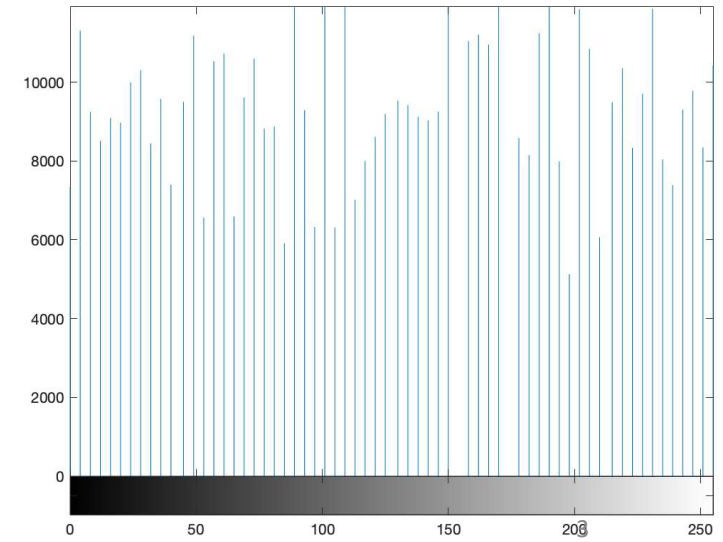
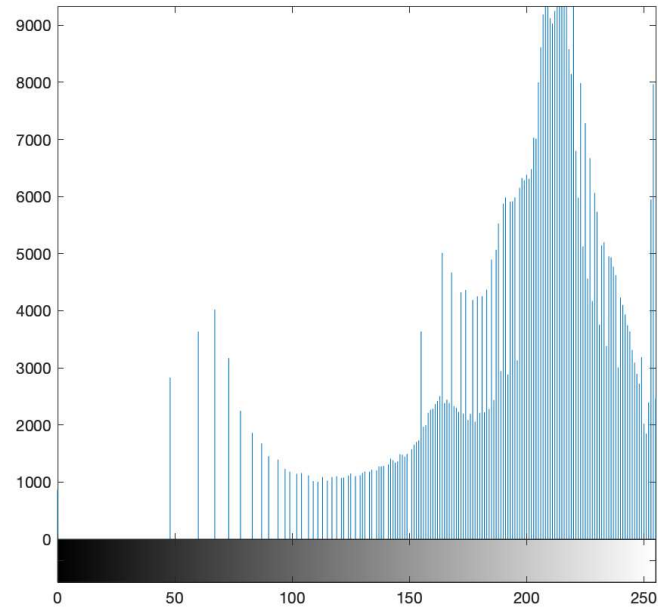
Histogram Equalization – Discrete Case

Probability p_k of occurrence of intensity level r_k is approximately $n_k/(MN)$ where:

- n_k is the number of pixels of that intensity
- MN is the total number of pixels in the image

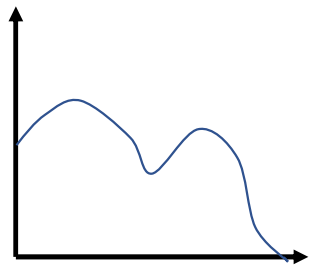
Then intensity r_k should get remapped to:

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_j$$

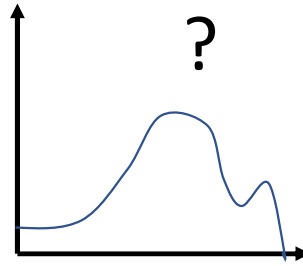


Histogram is not flat
It is “flat on the average”

Histogram Specification

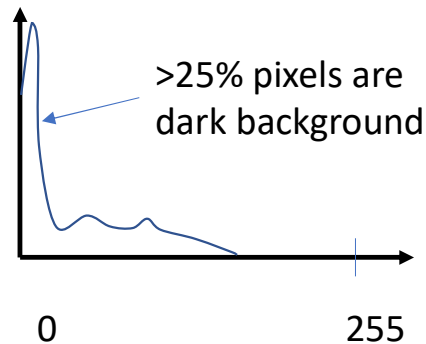
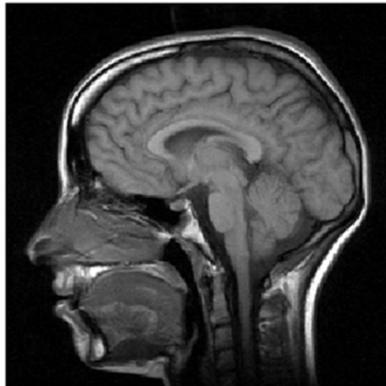


Input Histogram



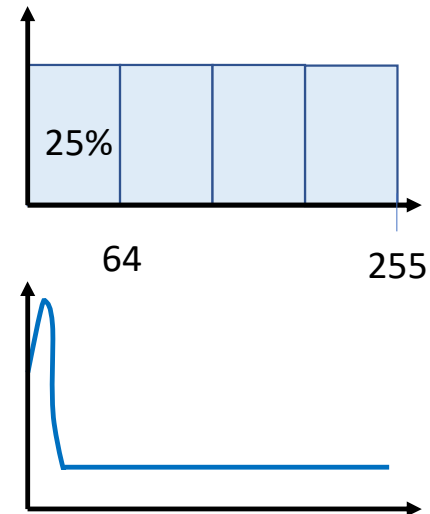
Desired Output Histogram

- How would you know what the output histogram should look like?
- You have a narrow class of images with similar characteristics



regular HE

hist. spec.

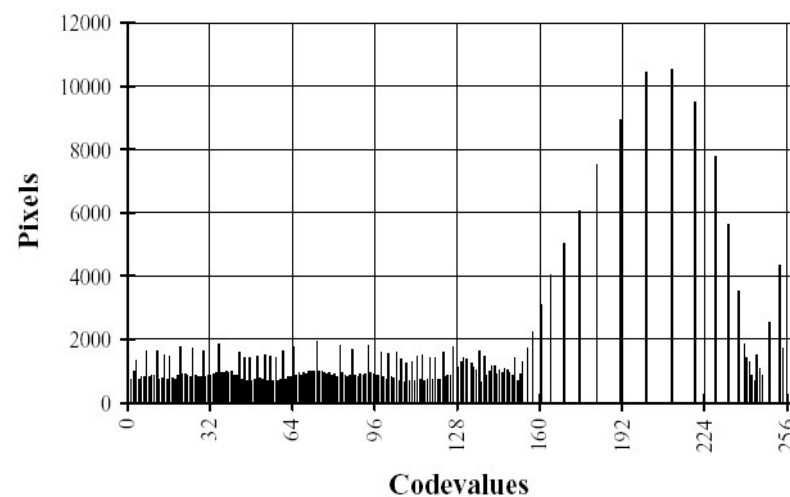
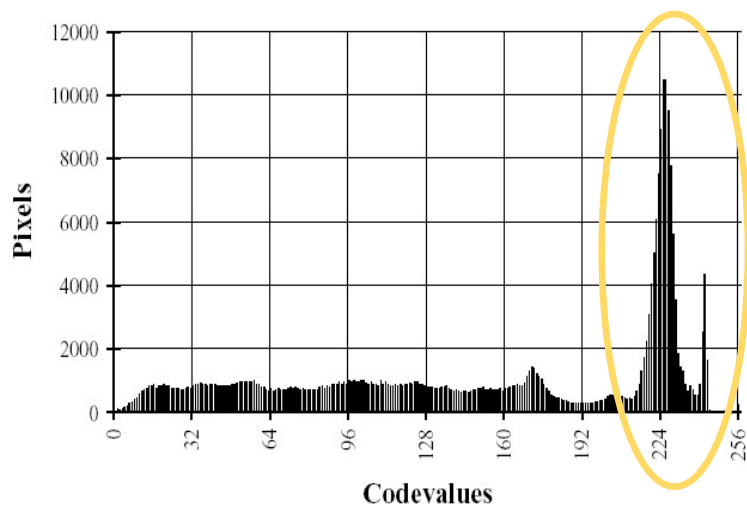


Actually, kind of a stupid way of dealing with this!

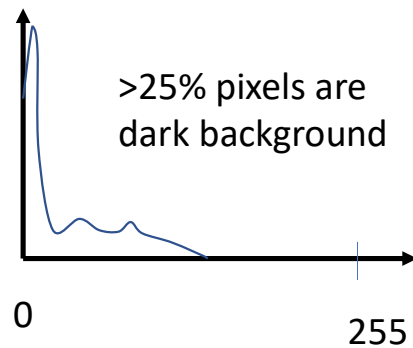
You have a
specific image,
but class is
not known



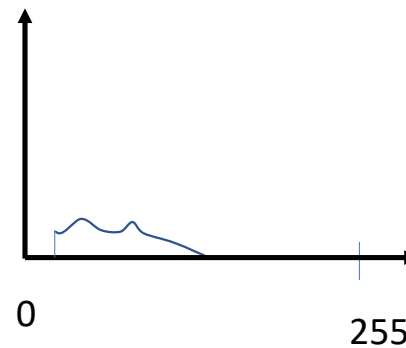
Problem



Can Modify the Input Histogram



zero out the
background pixels

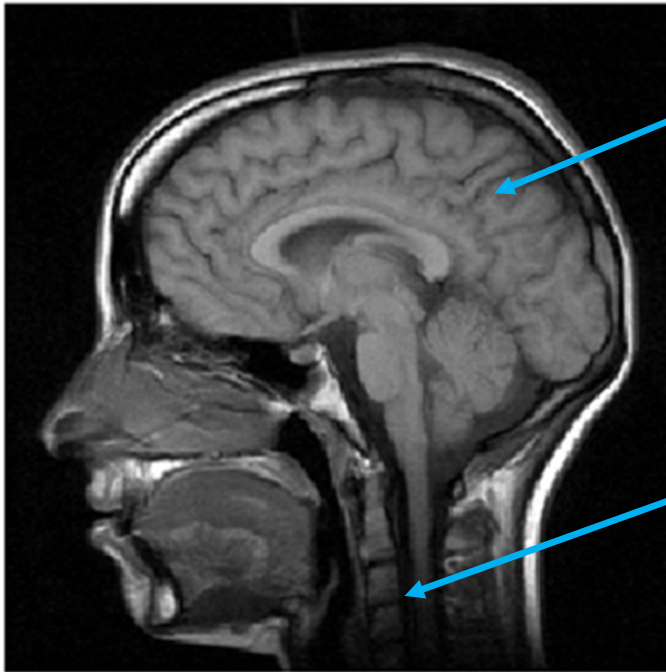


make CDF, apply
it to all pixels



- What happen to the pixels in the head?
- What happens to the pixels in the dark background?

Another Problem with Global Histogram Equalization



Cortex:

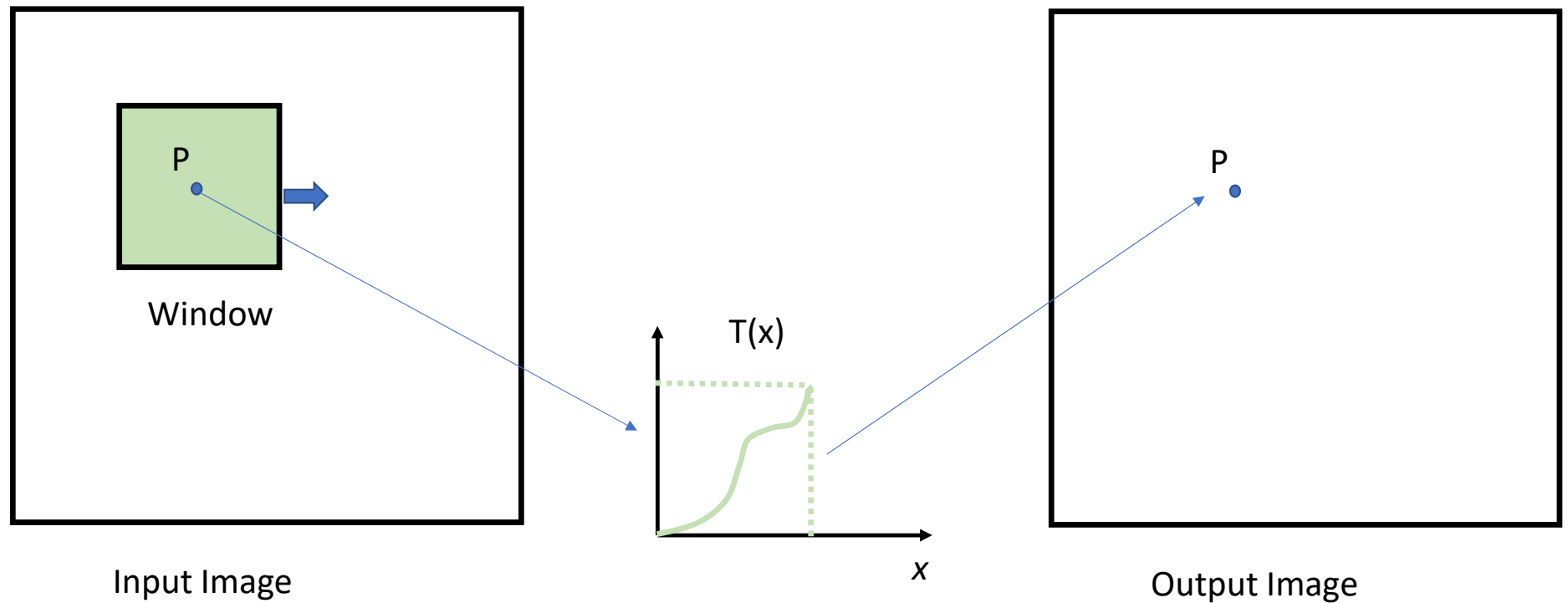
- Bright parts: 110-120, want to map them higher
- Dark parts: 70-80, want to map them lower

Vertebrae:

- Bright parts: 70-80, want to map them higher
- Dark parts: 30-40, want to map them lower

Adaptive Histogram Equalization (AHE)

Pointwise AHE: Pixel P is remapped using a remapping function that is the CDF of its own local window

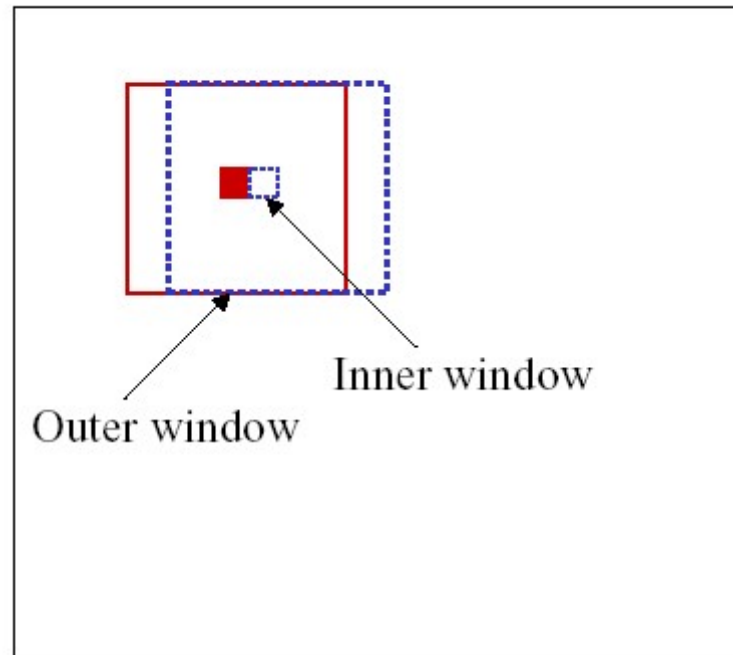


Pointwise AHE is complex

Moving window AHE: Small window is remapped using a remapping function that is the CDF of its own local larger window

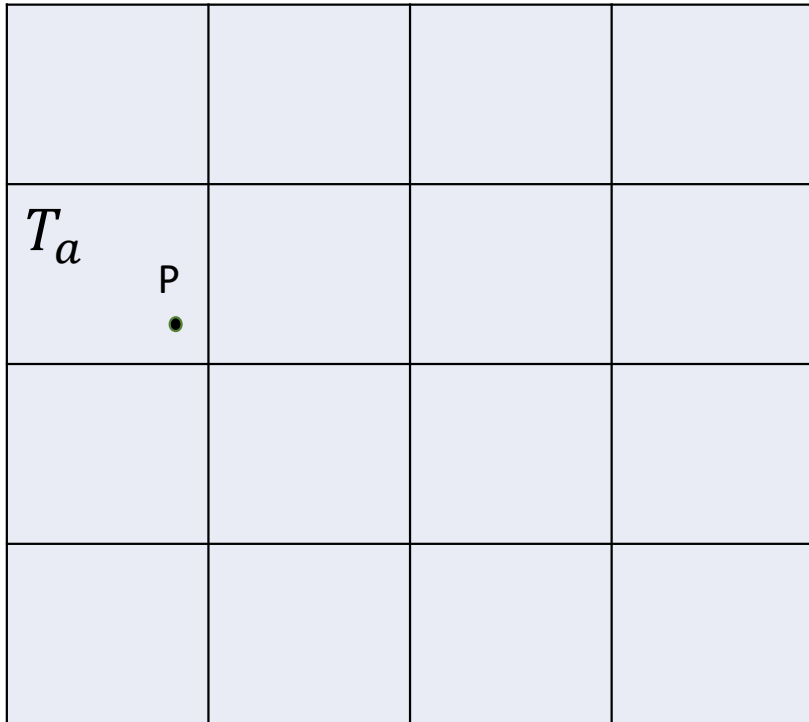
For example, the remapping function is:

- created from all pixels in a block of size 128 x 128
- then applied to small inner window of 16x16

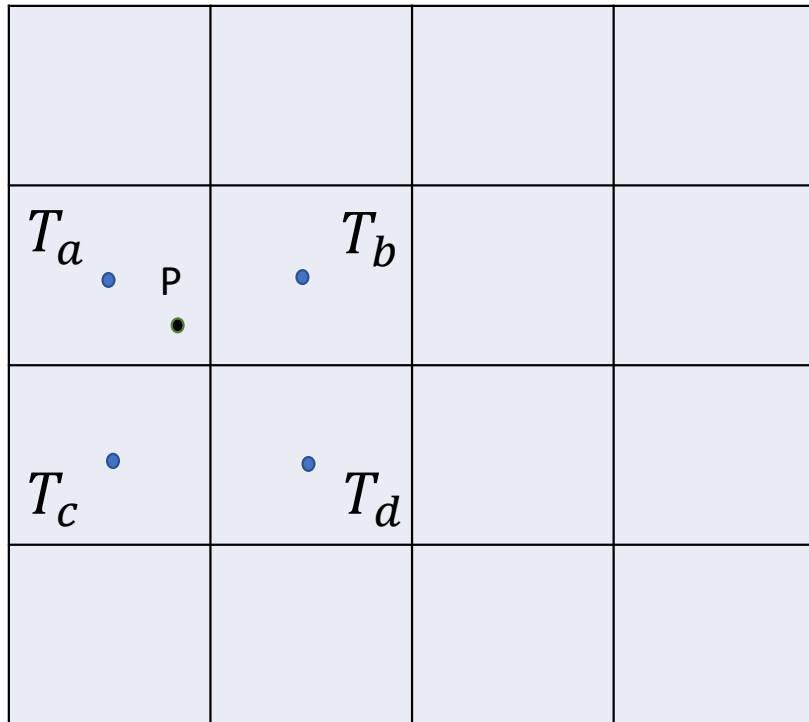


Can get even less complex

Compute a small number of remapping functions,
one for each block:



Block-based AHE: Need to combine blocks



Pixel P gets remapped using a weighted combination of the remapping functions from the 4 nearest blocks:

$$T_P = aT_a + bT_b + cT_c + dT_d$$

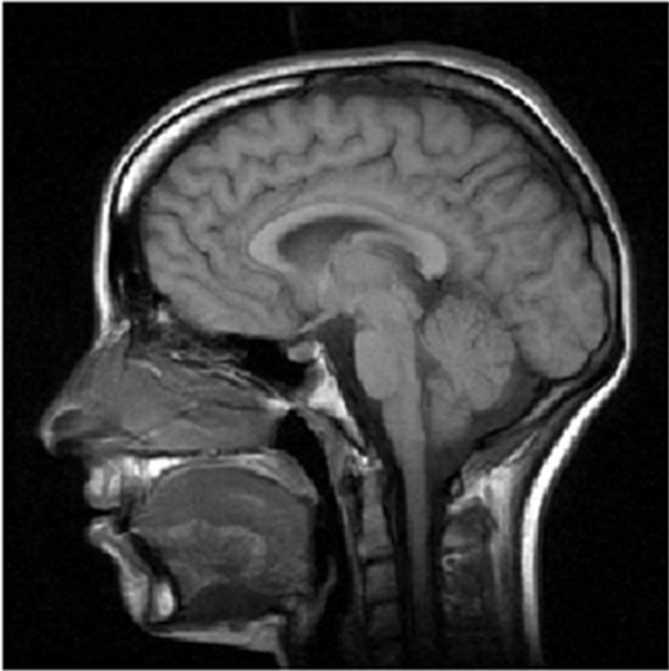
In Matlab, function is called *adapthisteq*

The weights come from Bilinear Interpolation

AHE Example



AHE Brain scan example



Narrow histograms for some blocks

→ remapping function can give huge contrast boosts

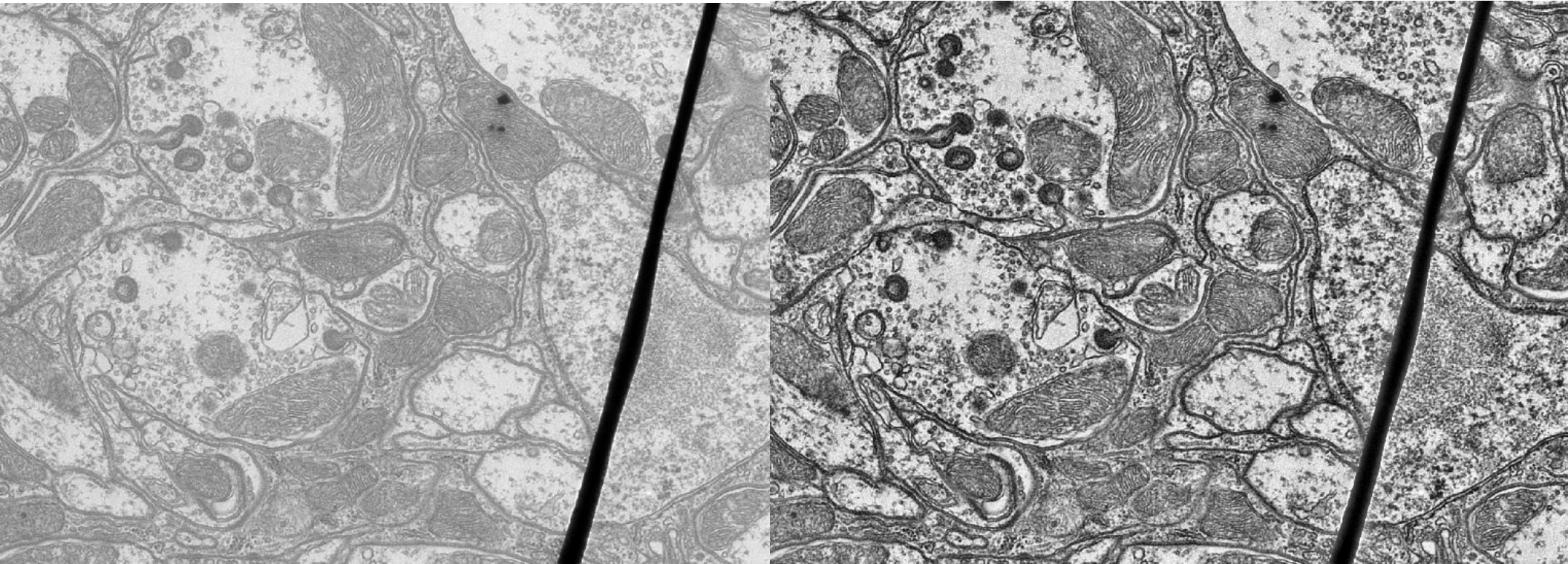
→ need contrast limiting

→ called CLAHE
Contrast Limited
Adaptive Histogram
Equalization)

In Matlab, *adapthisteq* implements AHE with bilinear interpolation + CLAHE

Need to control parameters to get what you want

CLAHE



TEM original image

TEM CLAHE processed
block: 50; bins: 256; max slope: 2.5

<https://imagej.net/plugins/clahe>

CLAHE

Photo1 original image



Photo1 CLAHE processed



block: 150; bins: 256; max slope: 3

Photo2 original image



Photo2 CLAHE processed



block: 50; bins: 256; max slope: 3



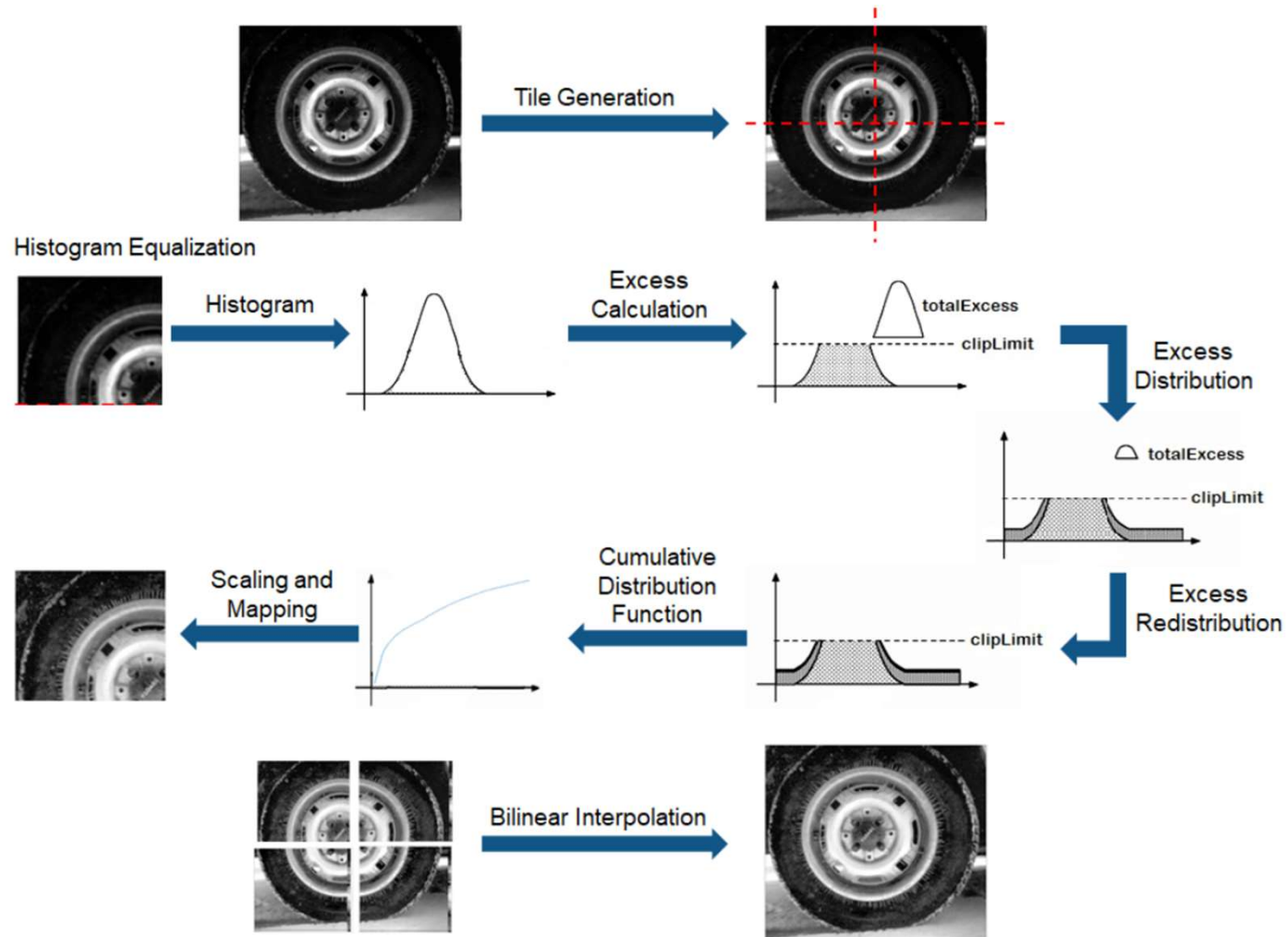


GHE

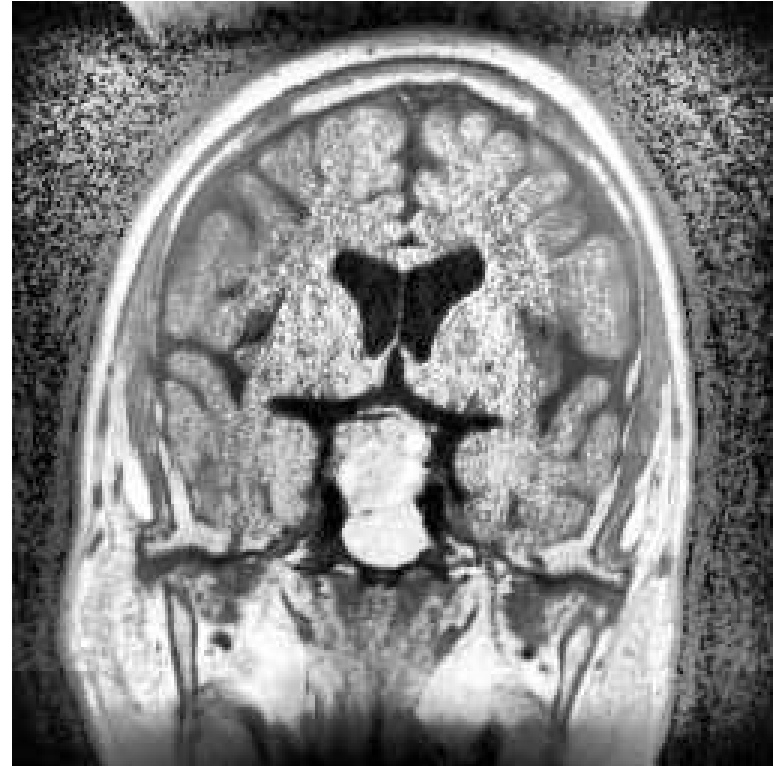
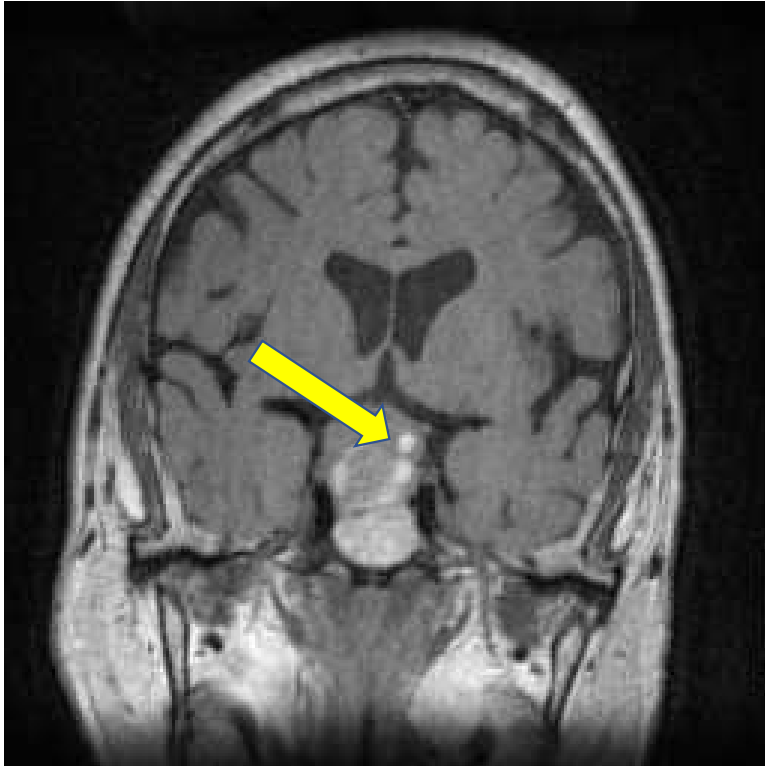


CLAHE

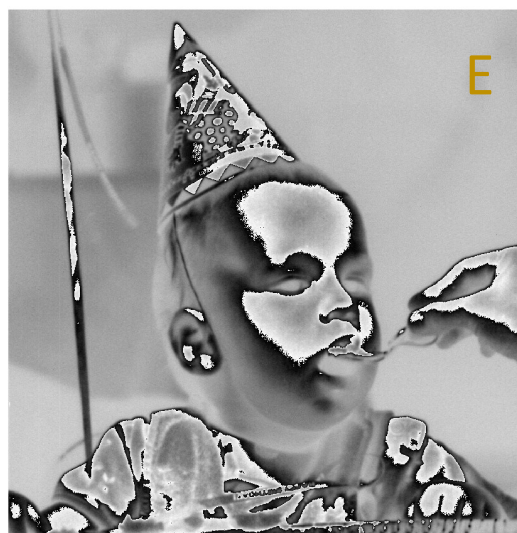
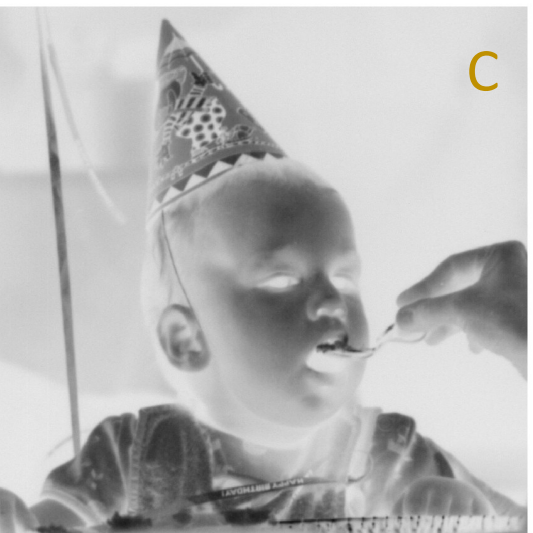
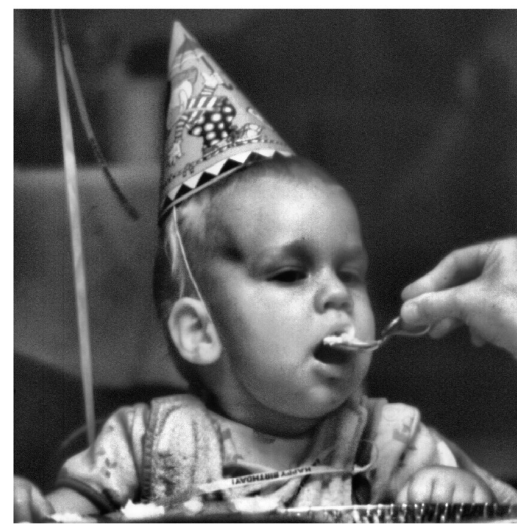
CLAHE



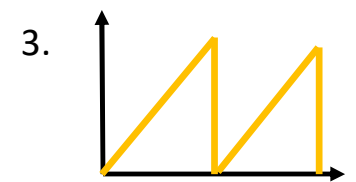
Bad Histogram Equalization



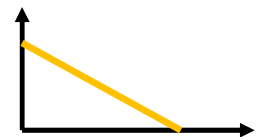
Reduced contrast on the small brain tumors,
which are the only things we care about seeing



1. GHE
2. AHE (*adapthisteq*)



4. Image negative



5. Windows & Levels

